



Shenzhen Hi-Link Electronics Co.,Ltd.

HLK-ZW0922 Specification
Round belt MCU Fingerprint Module



Content

1. Product introduction	1
2. Parameter	1
2.1. Technical indicators	1
2.2. Mechanical characteristics	2
2.3. Dimensions figure	2
3. Electrical parameters	2
4. Interface definition	3
4.1. Communication Interface	3
4.2. Connector type	3
4.3. Pin description	3
4.4. Actual photos	4
4.5. Fingerprint module internal circuit description	4
5. Version revision history	8

1. Product introduction

Consumers use fingerprint recognition modules for user identity determination. When the user touches the fingerprint recognition module with his finger, the fingerprint recognition module will scan the user's fingerprint, and the algorithm chip can obtain the fingerprint image data for registration and comparison.

2. Parameter

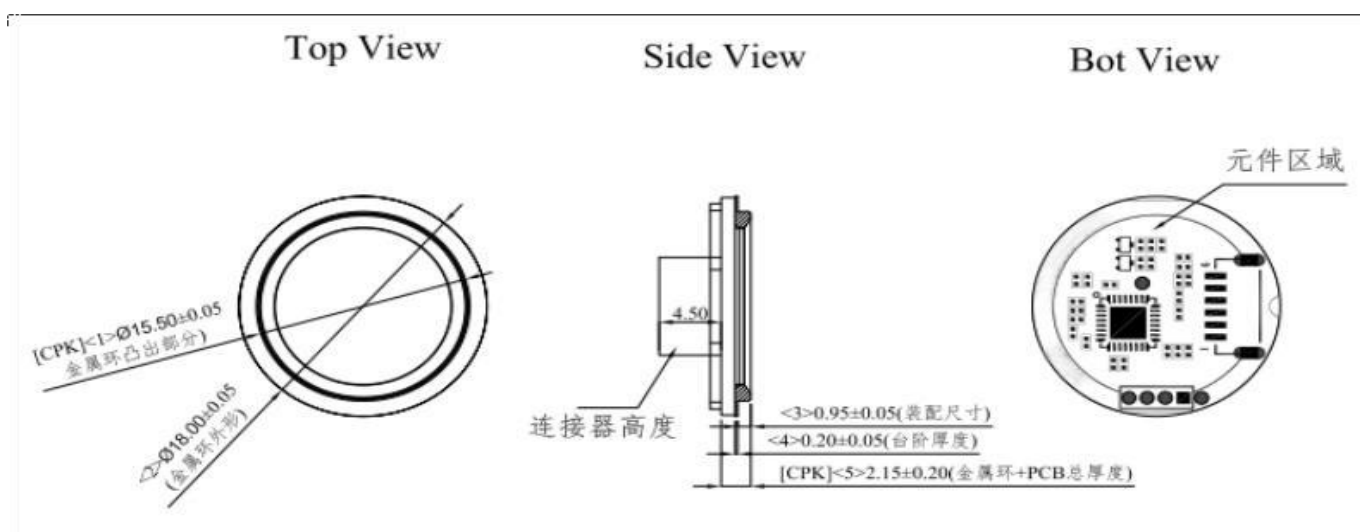
2.1. Technical indicators

Item	Parameter
Storage	50 pieces
Sensor type	Capacitive touch sensor
Resolution	508DPI
FRR.	<3%
FAR.	<0.001%
Fingerprint sensor shape	Round
Image gray level	8-bit grayscale
Service life	A million times
Static electricity test	Contact discharge $\pm 8KV$ /Air discharge $\pm 15KV$
Data interface	UART (TTL level)
Sensor surface hardness	3H
Working temperature and humidity range	-25°C~+70°C; 45%~85%RH
Storage temperature and humidity range	-40°C~+85°C; 45%~95%RH

2.2.Mechanical characteristics

Project	Description	Unit
Dimensions	Φ12.8mm	mm
Ribbon	5.6*4.8	mm
Sensor size	112*96	

2.3.Dimensions figure



3. Electrical parameters

Item		Min	Typical	Max	Unit
Supply voltage		3.0	3.3	3.6	V
Standby current (sensor)		8	10	12	μA
Operating current (algorithm MCU)		-	15	25	mA
Picture collection time		100	120	150	ms
Generate feature value time		60	80	100	ms
Comparison time		200	700	1300	ms
Power-on start time		-	100	-	ms
ESD rating	Non-contact discharge	-	15	-	KV
	Contact discharge	-	8	-	KV

4. Interface definition

4.1.Communication Interface

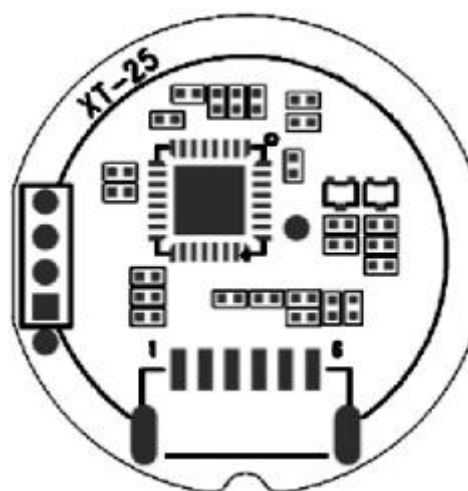
The standard UART default baud rate is 57600 bps, 1 start bit, 1 stop bit, 8 data bits, no parity bit, 3.3V TTL level.

4.2.Connector type

XH-1.00-6P: 6Pin 6-pin vertical connector, pitch 1.00mm.

4.3.Pin description

PIN	Define
1	SENSOR_3.3V
2	WAKEUP
3	MCU_3.3V
4	TX
5	RX
6	GND



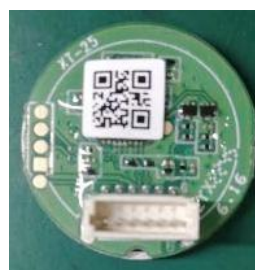
Pin number	Name	Type	Remark
1	SENSOR_3.3V	P	Sensor power supply, 3.3V
2	WAKEUP	O	Interrupt wake-up signal, (active high level)
3	MCU_3.3V	P	Algorithm chip power supply, 3.3V
4	TX	O	Serial data output
5	RX	I	Serial data input

6	GND	P	Ground signal
---	-----	---	---------------

Illustration:

- The serial port is a TTL level of 3.3V. If you need to connect to the serial port of PC, you need to connect the TTL-USB adapter board before communication.
- The SENSOR module circuit power supply needs to be powered at all times. Please ensure that the power supply has small ripple and is not interfered by other power supplies.
- If the lock board side performs a power-on and power-off operation for the fingerprint module; it is recommended that the power-off time interval be at least greater than 150ms.
- The master controller should follow the following sequence when controlling the fingerprint module MCU_3.3V to power on; Configure the serial port working mode after MCU_3.3V is powered on. It is not recommended to open the serial port first and then control MCU_3.3V to power on.
- After the main control goes to sleep, RX and TX should be pulled down to avoid excessive power consumption caused by leakage between IO ports.

4.4.Actual photos



4.5.Fingerprint module internal circuit description

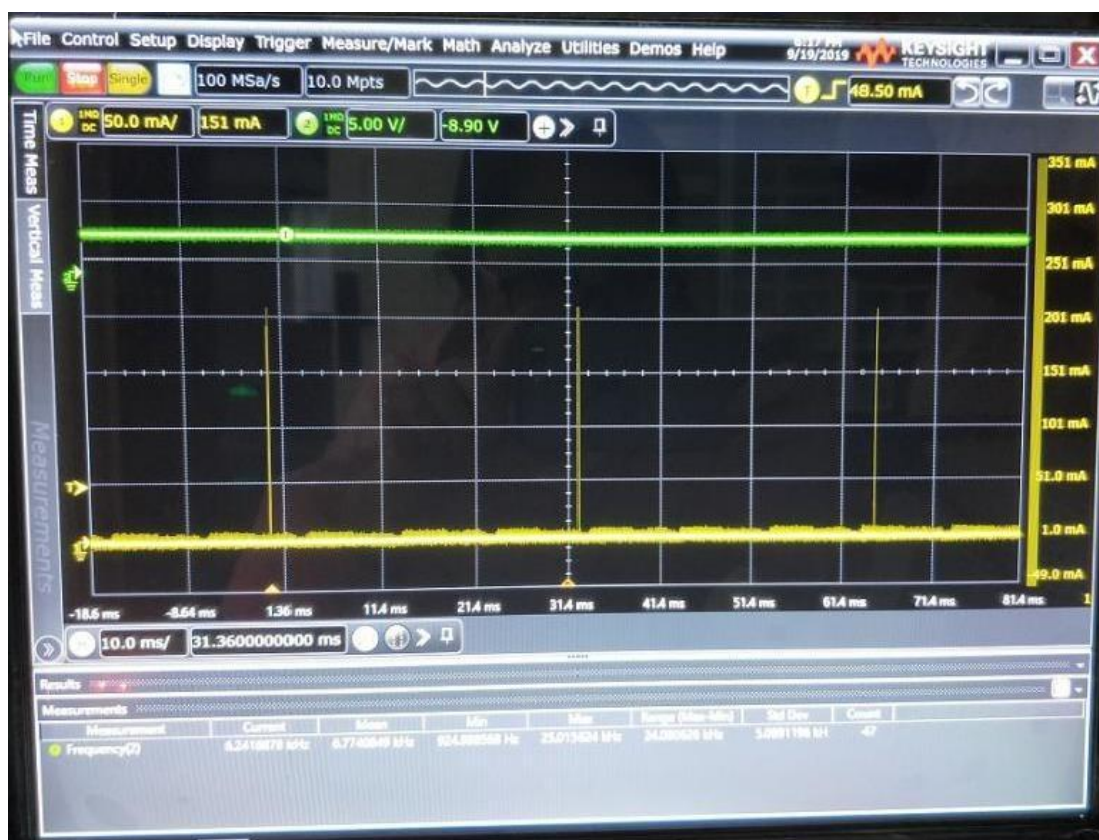
- Directly using the fingerprint sensor FD mode, the SENSOR wake-up is highly reliable and reduces the risk of false triggering.
- When there is no finger touch, the WAKEUP signal line is low level. When the finger touches the fingerprint sensor, the signal is triggered to high level, and then changes to low level when the finger leaves.

➤ **Fingerprint module workflow description:**

When the fingerprint module is in sleep state, WAKEUP is used as a wake-up signal. When the finger touches the fingerprint sensor, the interrupt signal is triggered, and then wakes up the system. The main control MCU controls the power-on of MCU_3.3V. After the fingerprint module algorithm chip is powered on and initialized, the main control MCU controls the fingerprint module through serial port commands to complete image collection and registration. , comparison task. After the task is completed, the main control chip controls the fingerprint module to sleep and controls the MCU_3.3V to power off. The system enters sleep and waits for the next round of work. After the system wakes up, the main control MCU can mask the interrupt signal, and turn on interrupt detection again before going to sleep after the task is completed.

➤ **Fingerprint sensor chip power supply requirements:**

The characteristics of the fingerprint chip itself: in FD (FingerDetect) mode, that is, when the fingerprint sensor periodically scans and detects fingerprints, a peak current of about 200mA will appear for 4us. Therefore, there are strict requirements for the power supply of the fingerprint chip. sensor_3.3V needs to control the power supply ripple within 200mV to avoid excessive power supply ripple causing the fingerprint sensor to reset at low voltage, causing the fingerprint sensor to fail to work properly.



Working current of fingerprint chip in FD mode

- It is necessary to use an LDO module with fast transient response to power the fingerprint chip; it is recommended that the LDO output current for Sensor power supply be $\geq 250\text{mA}$, ripple $< 100\text{mV}$, and $\text{PSRR} > 60\text{dB}$.

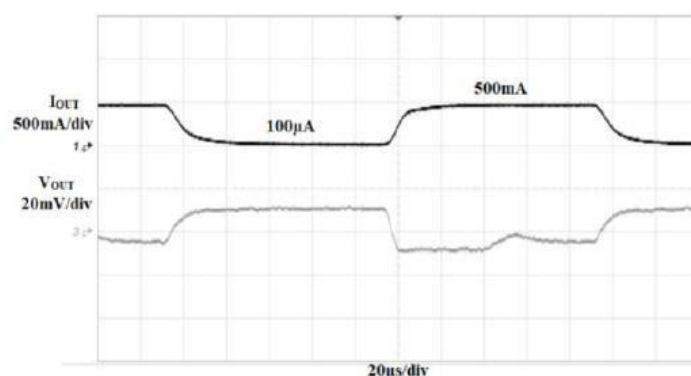
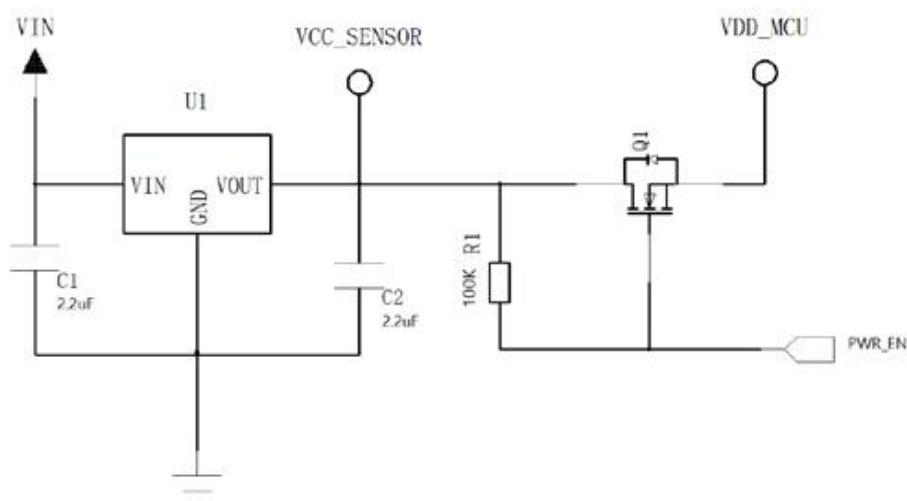
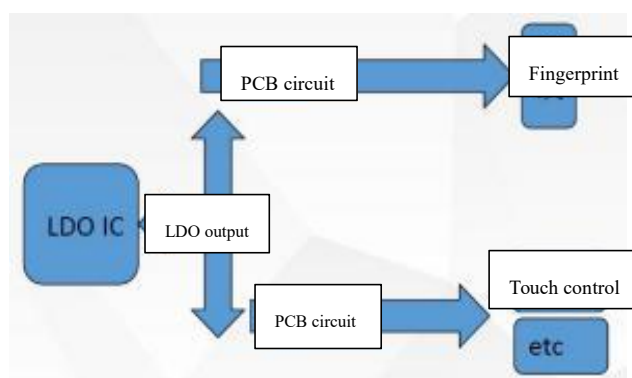


Figure 2.14 Load transient response waveform

- It is recommended to use a separate LDO to power the fingerprint module. The LDO output line supplies power to the fingerprint chip all the way, and a separate line is controlled by a MOS switch to power the fingerprint module algorithm MCU.



- When the fingerprint module shares the LDO power supply with other modules, the fingerprint module power supply needs to be routed separately from the LDO output on the PCB Layout.



5. Version revision history

Version	Revision Notes	date	Revised by
V1.0	first draft	2024-11-20	Chrales



Shenzhen Hi-Link Electronic Co., Ltd

Address: 17F,Building E, Xinghe WORLD, Minzhi Street, Long Hua district,
Shenzhen 51813

Email: sales@hlktech.com

Website: www.hlktech.net